Mathematical Scientific Computation
Modelling Biological Systems

Grade 9/10 Math Circles

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Modelling populations: Today's plan

1. Single-species population growth
2. Predator-prey populations
3. SIR model in epidemiology

Modelling single-species population



Let P_N be the penguin population in year N

$$\text{next year's population: } P_{N+1} = rP_N + m$$

- r is the **reproduction rate**
- m is the **migration rate**

Example: Modelling single-species population

$$P_{N+1} = rP_N + m$$

Suppose that:

- initial population $P_0 = 100$
- reproduction rate $r = 1.1$
- migration rate $m = 5$

$$\begin{aligned} P_1 &= 1.1P_0 + m \\ &= 1.1 \times 100 + 5 \\ &= 110 + 5 \\ &= 115 \end{aligned}$$

Exercise. Compute the next population value P_2.

Single-species modelling in python

We'll write a function that computes populations over time using

$$P_{N+1} = rP_N + m$$

for varying reproduction rates r

Predator-prey models



Let P_N and S_N be the population of penguins and seals in year N

next year's populations: $\begin{cases} P_{N+1} = r_p P_N + u S_N \\ S_{N+1} = v P_N + r_s S_N \end{cases}$

- r_p is **reproduction rate** for penguins, r_s is the reproduction rate for seals
- u and v are cross-species interaction parameters

Example: Predator-prey model

$$\begin{cases} P_{N+1} = 1.1P_N - 0.2S_N \\ S_{N+1} = 0.1P_N + 0.8S_N \end{cases}$$

Suppose that we have initial populations $P_0 = 90$ and $S_0 = 30$

$$\begin{aligned} P_1 &= 1.1 \times 90 - 0.2 \times 30 \\ &= 99 - 6 \\ &= 93 \end{aligned}$$

$$\begin{aligned} S_1 &= 0.1 \times 90 + 0.8 \times 30 \\ &= 9 + 24 \\ &= 33 \end{aligned}$$

Exercise. Compute the next population values P_2 and S_2.

Predator-prey modelling in python

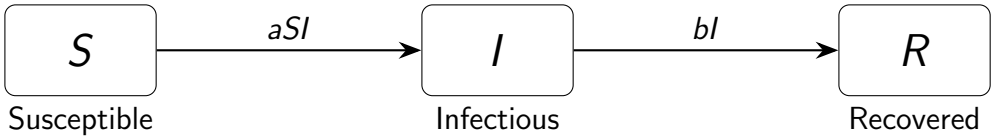
We'll write a function that computes populations over time using

$$\begin{cases} P_{N+1} = r_p P_N + u S_N \\ S_{N+1} = v P_N + r_s S_N \end{cases}$$

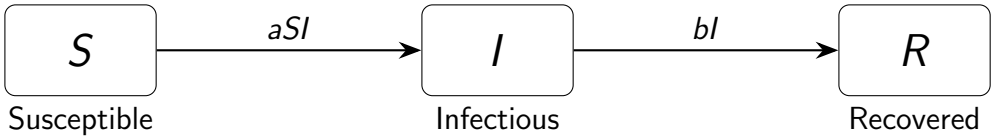
with parameters

$$P_0 = 90, \qquad r_p = 1.1, \qquad u = -0.2,$$
$$S_0 = 30, \qquad v = 0.1, \qquad s_p = 0.8.$$

SIR model for infectious disease



SIR model for infectious disease



$$\begin{cases} S_{N+1} = S_N - aS_N I_N \\ I_{N+1} = I_N + aS_N I_N - bI_N \\ R_{N+1} = R_N + bI_N \end{cases}$$

Example: SIR model

$$\begin{cases} S_{N+1} = S_N - aS_N I_N \\ I_{N+1} = I_N + aS_N I_N - bI_N \\ R_{N+1} = R_N + bI_N \end{cases}$$

Suppose that we have initial populations $S_0 = 990$, $I_0 = 10$, and $R_0 = 0$.

Take parameters $a = 0.0005$ and $b = 0.1$.

Example: SIR model

$$\begin{cases} S_{N+1} = S_N - aS_N I_N \\ I_{N+1} = I_N + aS_N I_N - bI_N \\ R_{N+1} = R_N + bI_N \end{cases}$$

Suppose that we have initial populations $S_0 = 990$, $I_0 = 10$, and $R_0 = 0$.

Take parameters $a = 0.0005$ and $b = 0.1$.

$$S_1 = 990 - 0.0005 \cdot 990 \cdot 10 \qquad = 990 - 4.95 \qquad = 985.05$$

Example: SIR model

$$\begin{cases} S_{N+1} = S_N - aS_N I_N \\ I_{N+1} = I_N + aS_N I_N - bI_N \\ R_{N+1} = R_N + bI_N \end{cases}$$

Suppose that we have initial populations $S_0 = 990$, $I_0 = 10$, and $R_0 = 0$.

Take parameters $a = 0.0005$ and $b = 0.1$.

$$S_1 = 990 - 0.0005 \cdot 990 \cdot 10 = 990 - 4.95 = 985.05$$
$$I_1 = 10 + 0.0005 \cdot 990 \cdot 10 - 0.1 \cdot 10 = 10 + 4.95 - 1 = 13.95$$

Example: SIR model

$$\begin{cases} S_{N+1} = S_N - aS_N I_N \\ I_{N+1} = I_N + aS_N I_N - bI_N \\ R_{N+1} = R_N + bI_N \end{cases}$$

Suppose that we have initial populations $S_0 = 990$, $I_0 = 10$, and $R_0 = 0$.

Take parameters $a = 0.0005$ and $b = 0.1$.

$$\begin{aligned} S_1 &= 990 - 0.0005 \cdot 990 \cdot 10 &&= 990 - 4.95 &&= 985.05 \\ I_1 &= 10 + 0.0005 \cdot 990 \cdot 10 - 0.1 \cdot 10 &&= 10 + 4.95 - 1 &&= 13.95 \\ R_1 &= 0 + 0.1 \cdot 10 &&= 0 + 1 &&= 1 \end{aligned}$$

Example: SIR model

$$\begin{cases} S_{N+1} = S_N - aS_N I_N \\ I_{N+1} = I_N + aS_N I_N - bI_N \\ R_{N+1} = R_N + bI_N \end{cases}$$

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$$\begin{aligned} S_1 &= 990 - 0.0005 \cdot 990 \cdot 10 &&= 990 - 4.95 &&= 985.05 \\ I_1 &= 10 + 0.0005 \cdot 990 \cdot 10 - 0.1 \cdot 10 &&= 10 + 4.95 - 1 &&= 13.95 \\ R_1 &= 0 + 0.1 \cdot 10 &&= 0 + 1 &&= 1 \end{aligned}$$

Exercise. Compute the next SIR values S_2, I_2, and R_2.
Check that $S_2 + I_2 + R_2 = 1000$.

SIR modelling in python

We'll write a function that computes populations over time using

$$\begin{cases} S_{N+1} = S_N - aS_N I_N \\ I_{N+1} = I_N + aS_N I_N - bI_N \\ R_{N+1} = R_N + bI_N \end{cases}$$

for varying transmission rates a, b